

Electronic Material Industry

Electronic material monthly: PCB materials, micro-thin foil, optics, wafers

Industry Overview

PCB materials: Glass cloth and copper foil prices rising

Combined July sales of the three major Taiwanese copper-clad laminate (CCL) manufacturers increased 122% YoY, maintaining strong momentum. Sales at Baotek, a subsidiary of Nittobo, were also firm. Market prices for electronic glass cloth and glass yarn in China continued to rise. Prices for Asahi Kasei's low-dielectric glass cloth and Nittobo's low-dielectric glass yarn and glass cloth also increased modestly. Copper foil prices continued to strengthen. During the earnings season, Mitsui Kinzoku reported price hikes for VSP foil, while Furukawa Electric benefited from copper foil price increases and a favorable product mix driven by higher volumes of HVLP4 foil. Overall, these are positive data points for PCB material suppliers.

Micro-thin foil: Optical transceiver demand rises

During 1Q earnings announcements, it was confirmed that demand for Mitsui Kinzoku's micro-thin foil increased significantly, driven by 800GbE optical transceiver applications. According to leading PCB manufacturer Compeq, mSAP processes are used in PCB production for 800GbE optical transceivers, with 4-8 of the 12-14 layers utilizing the mSAP process. For 1.6TbE optical transceivers, mSAP is applied to 6-10 of the 14-16 layers. Optical transceivers require greater usage of micro-thin foil. Together with increasing demand for memory packaging applications, this strengthens expectations for robust growth in micro-thin foil demand.

Optical fiber/devices: Demand stronger than expected

Optical fiber market conditions continued to improve in July. According to Fujikura, optical cable prices increased by 25% for a major customer project, contributing to the upward revision of guidance in its 1Q results. Furukawa Electric's strong earnings were also supported by the positive impact of optical cable price increases. For optical laser products, Lumentum and Coherent commented at their earnings briefings that there has been no delay in the adoption of CPO, inquiries continue to strengthen, NPO represents an additional market opportunity, product supply capacity remains constrained, and prices for certain products have increased. Furukawa Electric raised its outlook for optical device products in 1Q, supported in part by higher sales of SOAs, which carry higher margins than DFB chips. Overall, market conditions for optical cables and optical devices appear stronger than expected and remain positive for wire and cable manufacturers.

Semis materials: Strong July data for silicon wafers

Silicon wafer imports into Korea and Taiwan increased 24.9% YoY in July. In particular, silicon wafer imports into Korea rose 44% YoY, representing particularly strong growth.

(Continued on next page)

20 August 2026

Equity
Japan
Chemicals-Specialty

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Takashi Enomoto >>
Research Analyst
BofAS Japan
takashi.enomoto@bofa.com

Niima Komura >>
Research Analyst
BofAS Japan
niima.komura@bofa.com

Kyoji Chiba >>
Research Analyst
BofAS Japan
kyoji.chiba@bofa.com

Yuri Tanaka >>
Research Analyst
BofAS Japan
yuri.tanaka@bofa.com

CPO: Co-Packaged Optics
DFB: Distributed Feedback laser
HVLP: Hyper Very Low Profile
mSAP: Modified Semi-Additive Process

NPO: Near-Packaged Optics

PCB: Printed Circuit Board

SOA: Semiconductor Optical Amplifier

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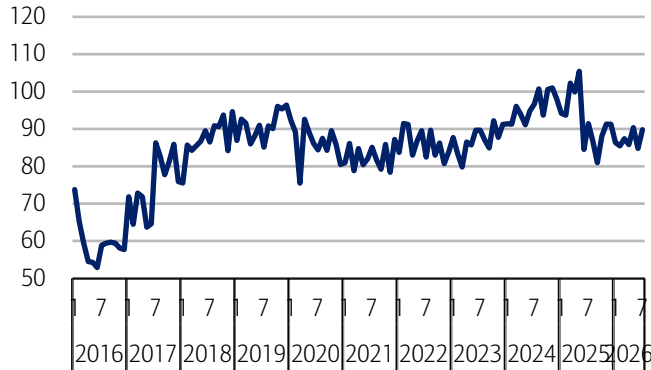
Refer to important disclosures on page 14 to 16. Analyst Certification on page 12. Price Objective Basis/Risk on page 11.

Shin-Etsu Chemical commented at its briefing that July shipments of 300mm wafers reached a record monthly high. While this was not a surprise, it remains a positive data point. Formosa SUMCO's July sales increased 28.8% YoY. According to SUMCO, price pass-through is becoming visible in the spot market for silicon wafers used in applications where demand is increasing, but not yet for applications where demand has yet to recover.

Semiconductor materials

Exhibit 1: Unit prices increased MoM

300mm wafer estimated import price in Taiwan

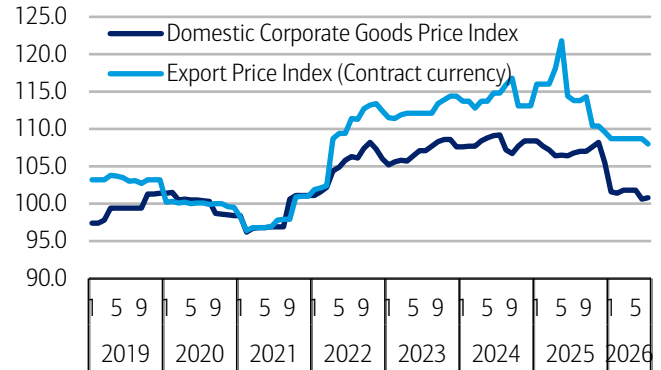


Source: BofA Global Research estimates from Taiwan custom

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Exhibit 2: Flat MoM

Japan corporate goods price index of silicon wafer

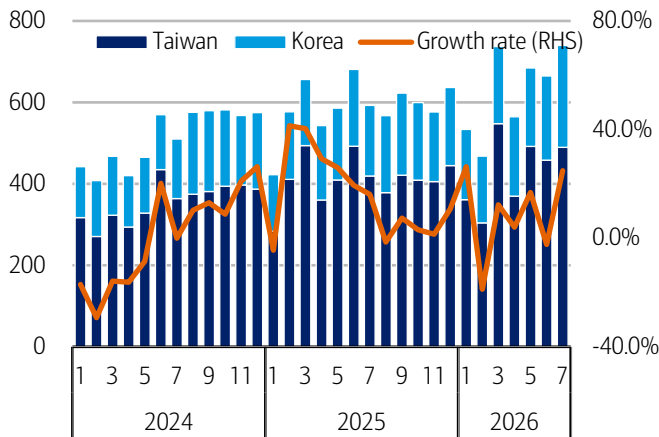


Source: Bank of Japan

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Exhibit 3: July increased 24.9% YoY

Silicon wafer import volume of Taiwan and Korea (ton, YoY)

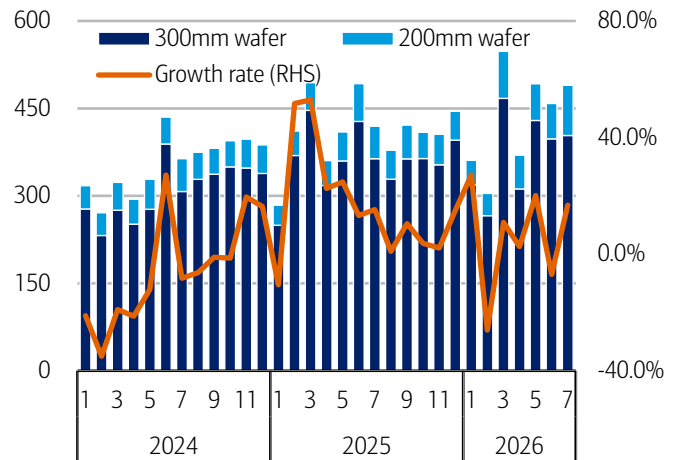


Note: Taiwan silicon wafer import comprises 300mm and 200mm wafers. Source: Taiwan and Korea customs

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Exhibit 4: July increased 16.8% YoY

300mm and 200mm wafers import volume of Taiwan (ton, YoY)

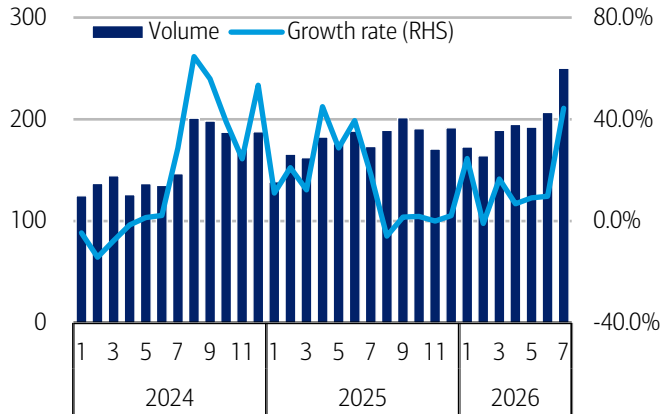


Source: Taiwan custom

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Exhibit 5: July increased 44.4% YoY

Silicon wafer import of Korea (ton, YoY)

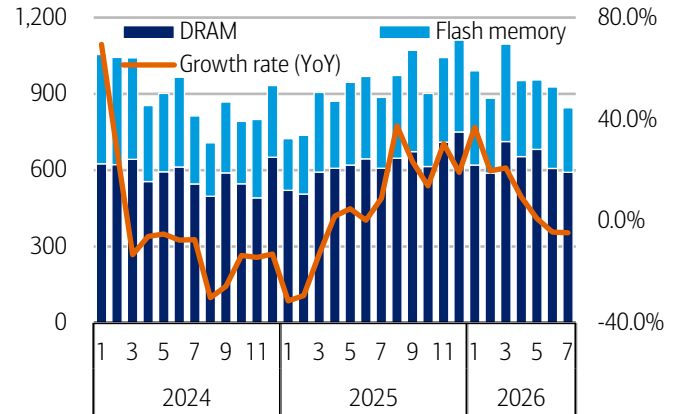


Source: Korea custom

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Exhibit 6: July decreased 4.6% YoY

DRAM and flash memory export of Korea (million units, YoY)

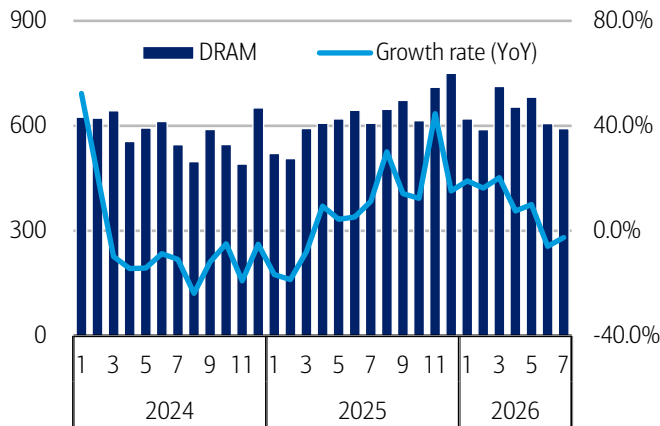


Source: Korea custom

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Exhibit 7: July export volumes decreased 2.6% YoY

DRAM export of Korea (million units, YoY)

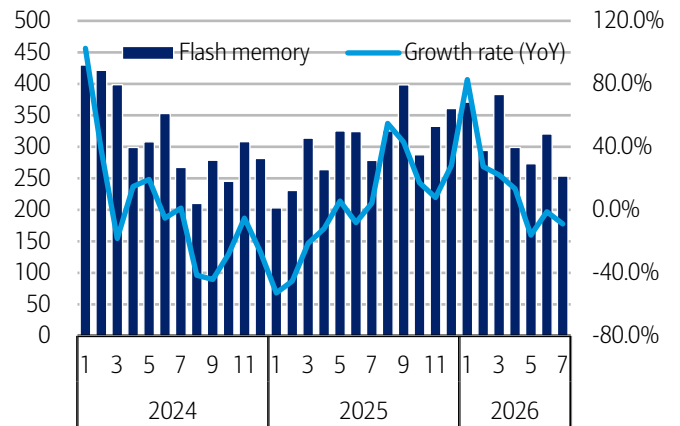


Source: Korea custom

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Exhibit 8: July decreased 9.0% YoY

Flash memory export of Korea (million units, YoY)

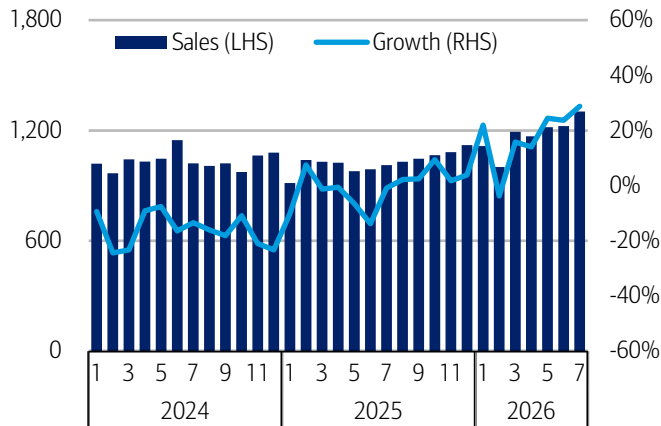


Source: Korea custom

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Exhibit 9: July increased 28.8% YoY

Formosa SUMCO Technology (3532 TT) monthly sales (million TW\$, YoY)

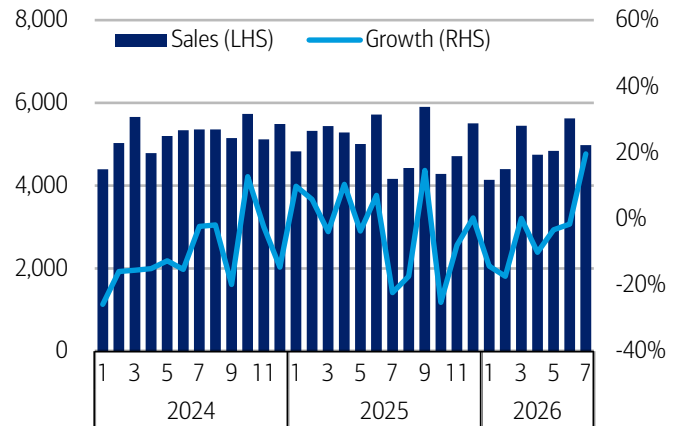


Source: Company, BofA Global Research

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Exhibit 10: July increased 19.6% YoY

Global Wafers (6488 TT) monthly sales (million TW\$, YoY)

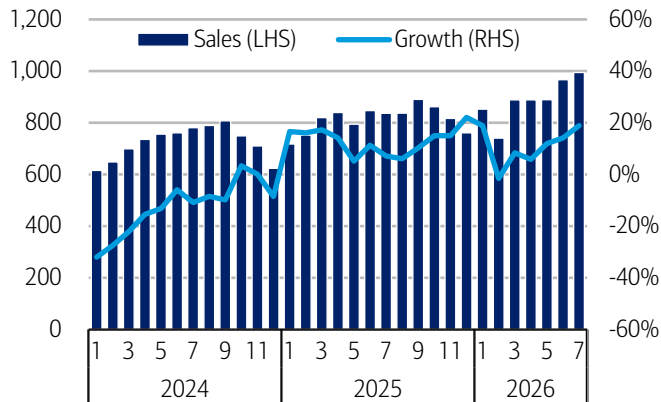


Source: Company, BofA Global Research

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Exhibit 11: July increased 18.9% YoY

Wafer Works (6182 TT) monthly sales (million TW\$, YoY)

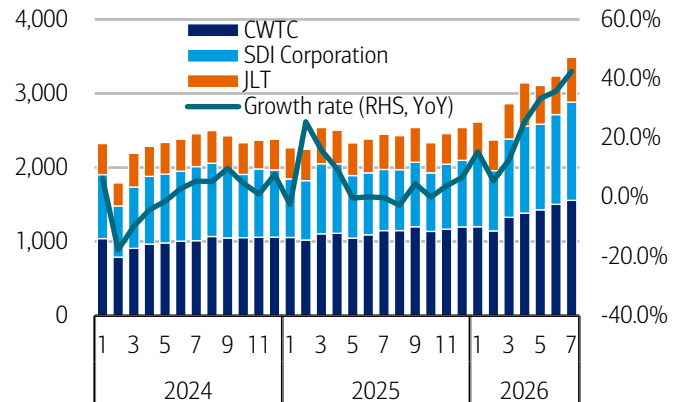


Source: Company, BofA Global Research

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Exhibit 12: July increased 42.4% YoY

Monthly sales of ceramic condenser players in Taiwan (Million Taiwan\$, YoY)

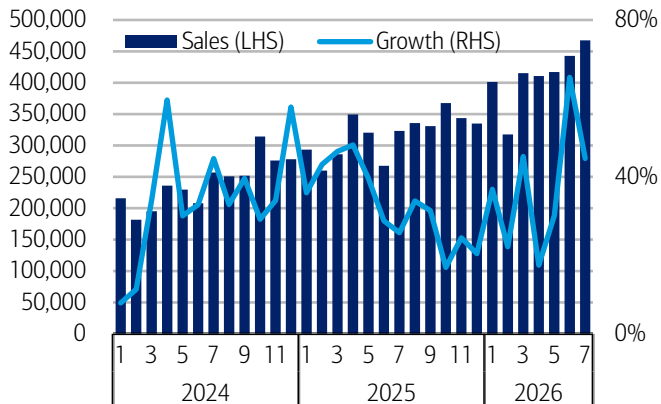


Source: MOPS

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Exhibit 13: July increased 44.7% YoY

TSMC (2330 TT) monthly sales (mn TW\$)

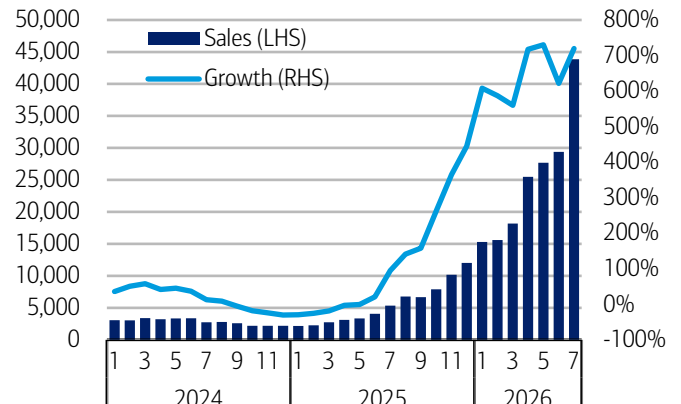


Source: Company, BofA Global Research

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Exhibit 14: July increased 719.6% YoY

Nanya Technology (2408 TT) monthly sales (mn TW\$)



Source: Company, BofA Global Research

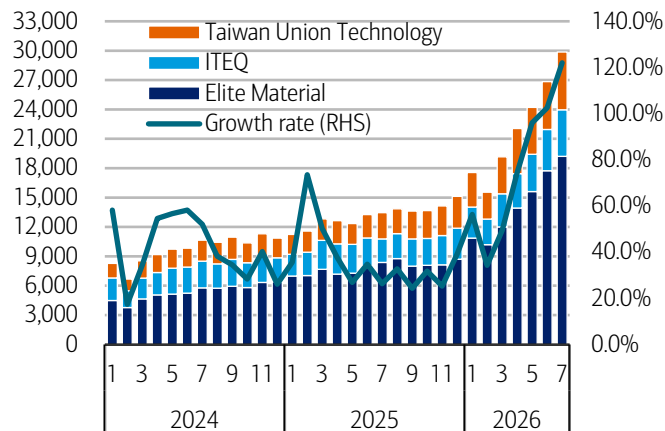
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Printed circuit board materials

Exhibit 15: July increased 122.0% YoY

Taiwan copper clad laminate (CCL) players' sales (million TW\$, YoY)

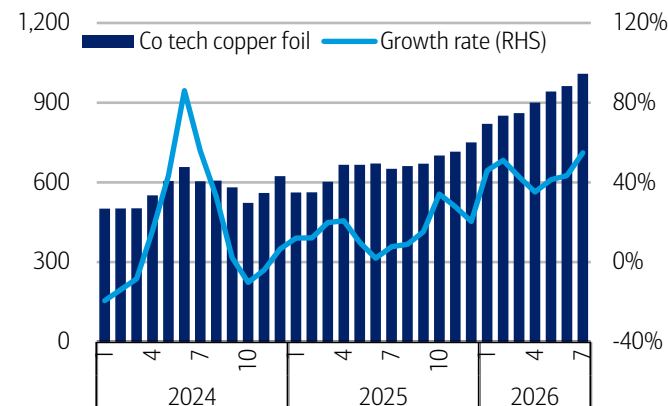


Source: MOPS

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Exhibit 16: July increased 54.9% YoY

Monthly sales of Co tech copper foil (Taiwan copper foil major player, million TW\$)

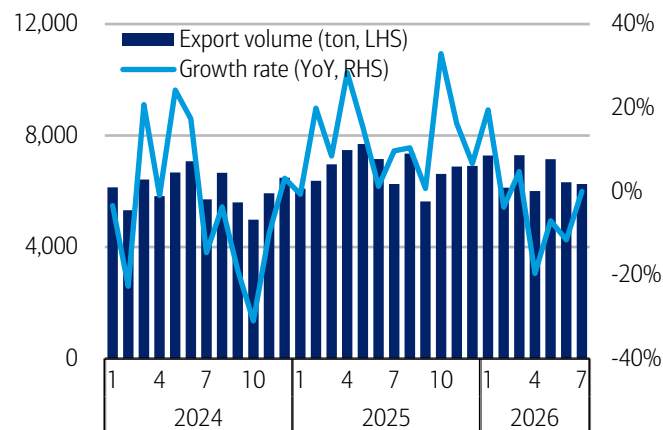


Source: MOPS

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Exhibit 17: July decreased 0.1% YoY

Taiwan copper foil export volume (less than 70micron)

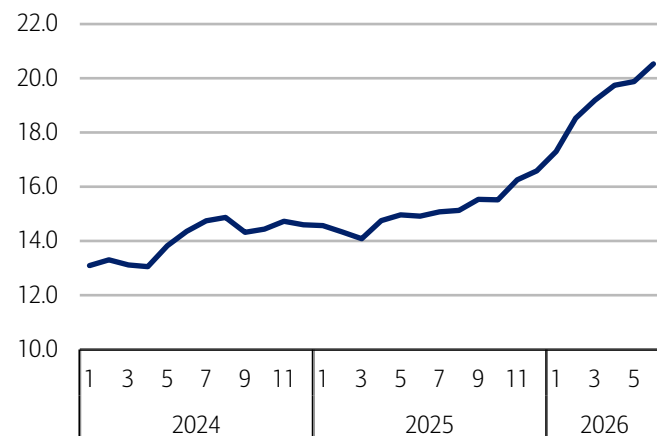


Source: Taiwan customs

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Exhibit 18: Export prices increased

Taiwan copper foil export price (US\$/kg)

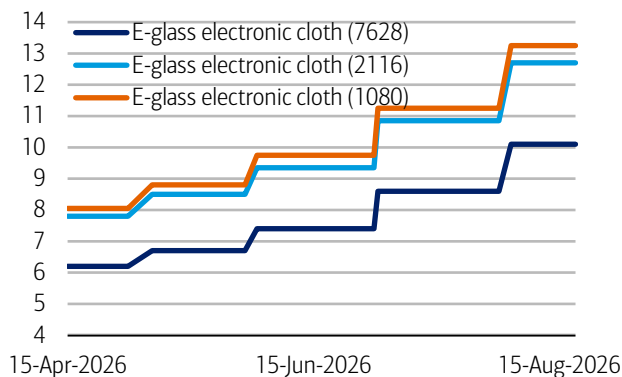


Source: Taiwan customs

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Exhibit 19: Prices are on an upward trend

China electronic cloth average prices (South China market, RMB/m)

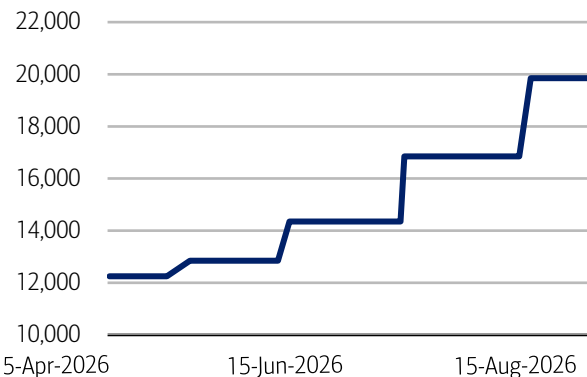


Source:SCI99

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Exhibit 20: Prices are on an upward trend

China electronic yarn average prices (East China market, RMB/ton)

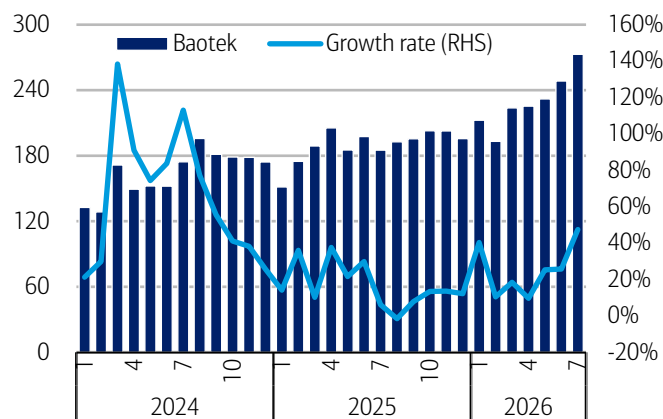


Source:SCI99

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Exhibit 21: Baotek is the subsidiary of Nittobo (3110)

Monthly sales of Baotek (Million Taiwan\$, YoY)

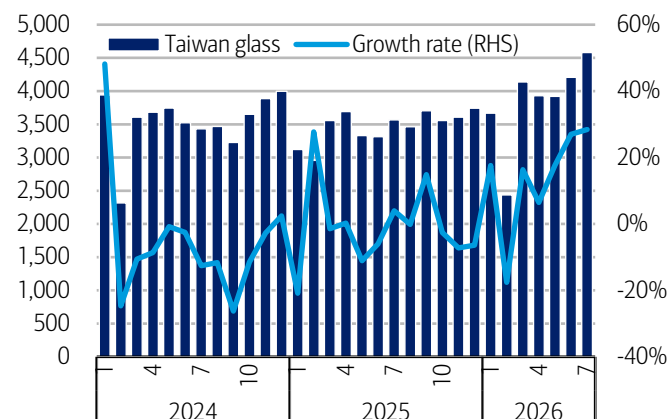


Source: MOPS

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Exhibit 22: July increased 28.4% YoY

Monthly sales of Taiwan glass (million TW\$, YoY)

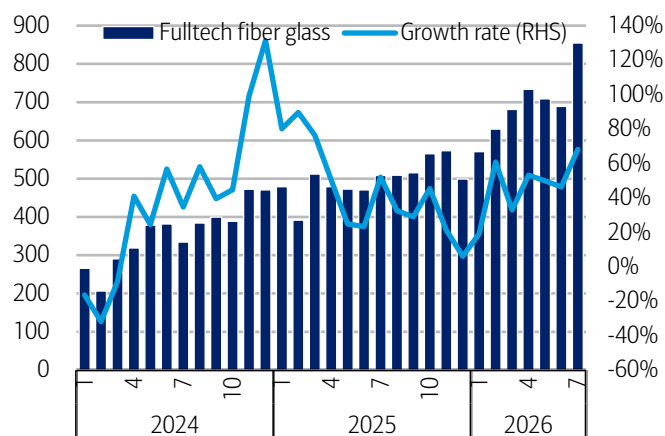


Source: MOPS

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Exhibit 23: July increased 68.1% YoY

Monthly sales of Fulltech fiber glass (million TW\$, YoY)

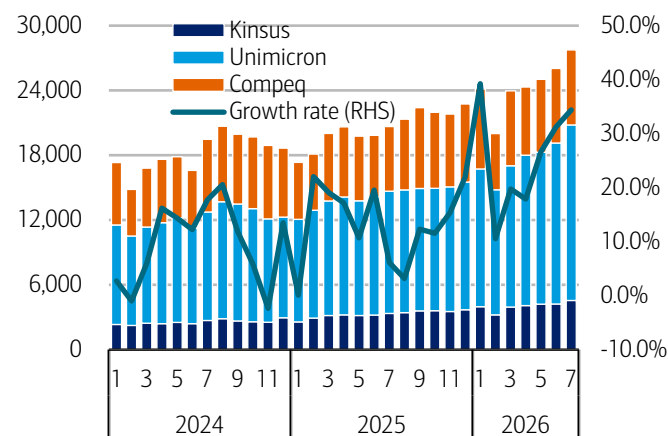


Source: MOPS

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Exhibit 24: July increased 34.4% YoY

Monthly sales of Taiwan-listed PCB suppliers (million TW\$, YoY)



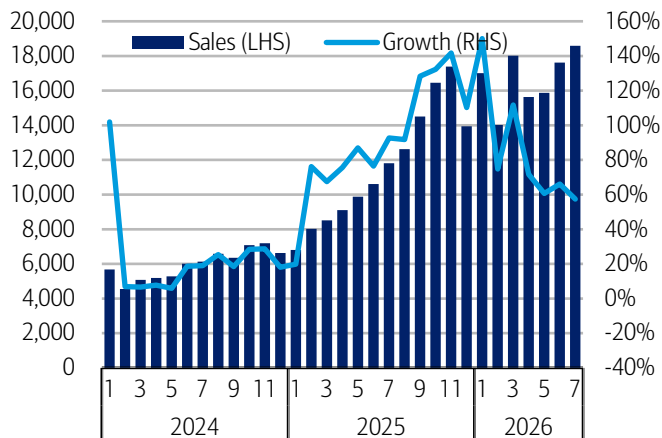
Source: MOPS

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Electronic components

Exhibit 25: July increased 57.4% YoY

Asia Vital Components (3017 TT) monthly sales (million TW\$, YoY)

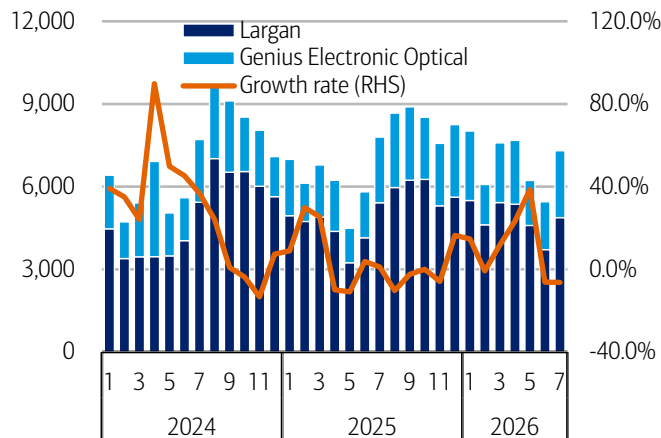


Source: Company, BofA Global Research

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Exhibit 26: July decreased 6.4% YoY

Sales growth of Largan and Genius Electronic Optical (million TW\$, YoY)

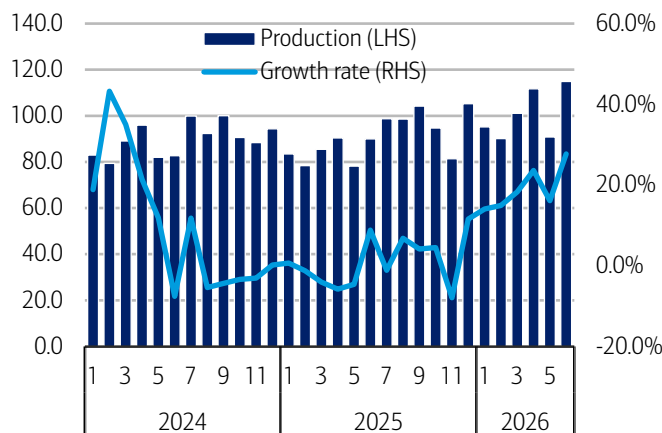


Source: MOPS, BofA Global Research

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Exhibit 27: Increased 27.7% YoY

Japan ceramic capacitor production volume and growth rate (bn units)

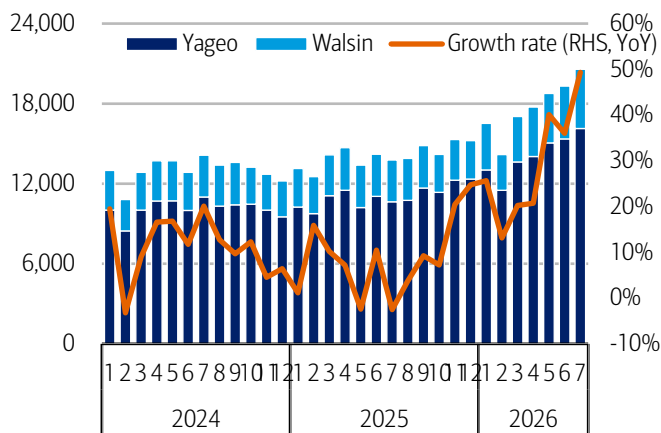


Source: METI (ministry of economy trade and industry)

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Exhibit 28: Increased 49.0% YoY

Monthly sales of ceramic condenser players in Taiwan (million TW\$, YoY)



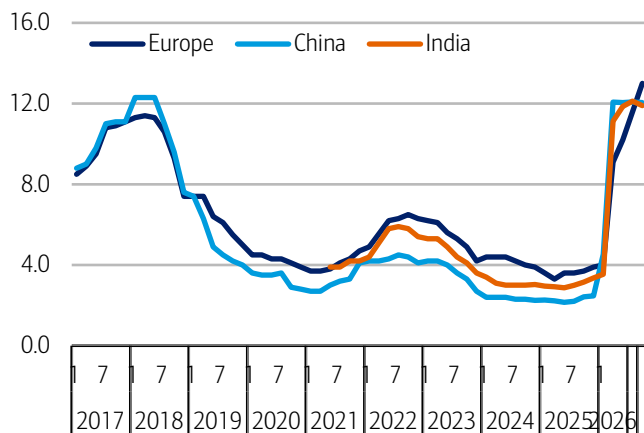
Source: MOPS

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Optical fiber, cables, and devices

Exhibit 29: Prices are trending upward

Price of G.652.D (standard single-mode fiber) (US\$/fiber-km)

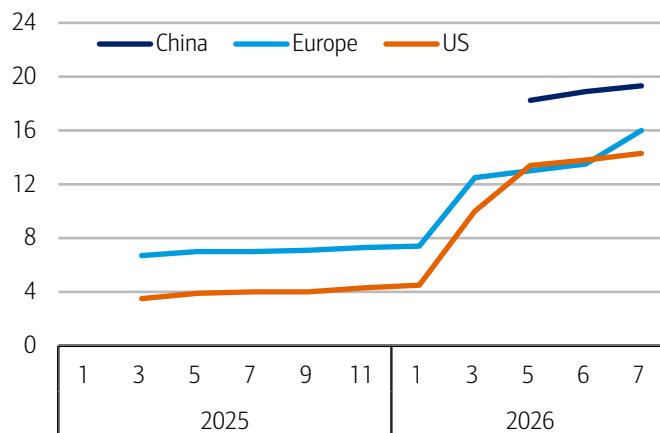


Source: CRU

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Exhibit 30: Prices remained flat

G.657.A1 (bend-insensitive single-mode fiber) price (US\$/fiber-km)

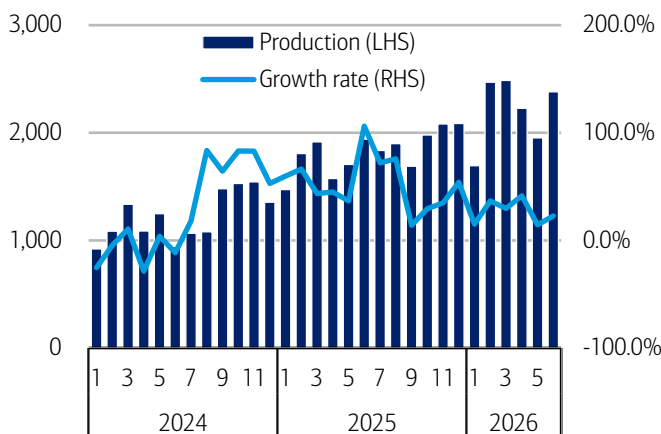


Source: CRU

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Exhibit 31: Increased 22.7% YoY

Japan optical fiber cable production volume and growth rate (000km core)

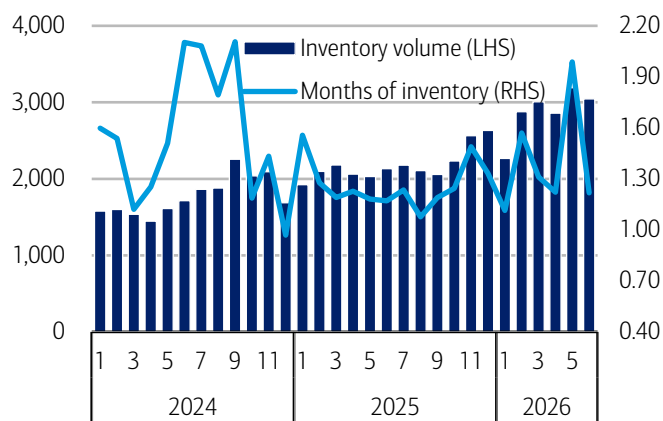


Source: METI (ministry of economy trade and industry)

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Exhibit 32: Inventories decreased MoM

Japan optical fiber cable inventory volume and months of inventory (000km core)

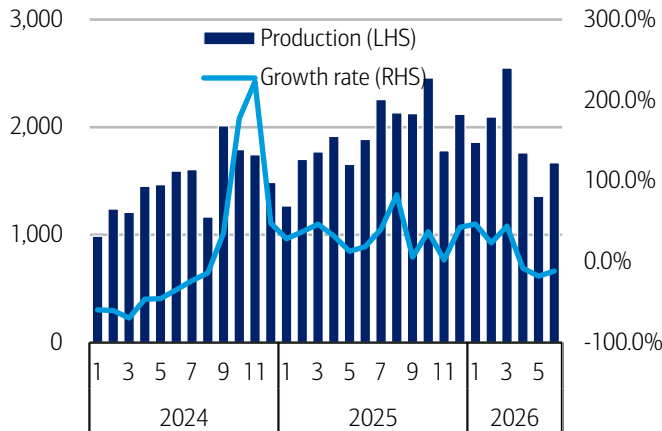


Source: METI (ministry of economy trade and industry)

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Exhibit 33: Decreased 11.6% YoY

Japan optical fiber production volume and growth rate (000km core)

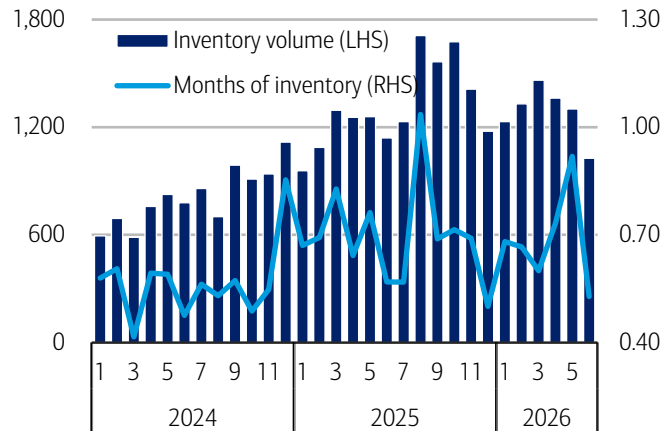


Source: METI (ministry of economy trade and industry)

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Exhibit 34: Inventories decreased MoM

Japan optical fiber production volume and growth rate (000km core)

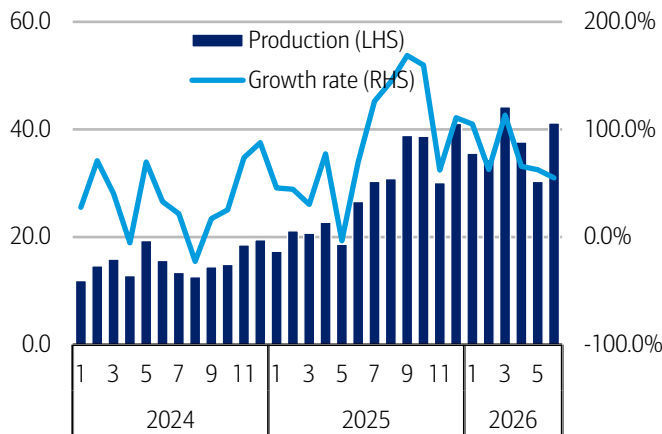


Source: METI (ministry of economy trade and industry)

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Exhibit 35: Increased 54.9% YoY

Japan laser diode production volume and growth rate (mn units)

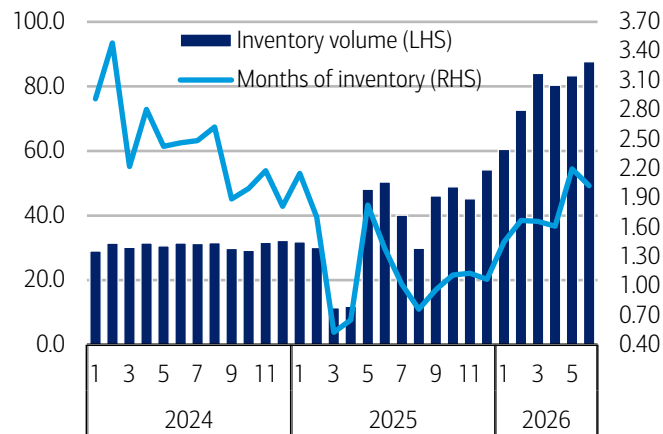


Source: METI (ministry of economy trade and industry)

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Exhibit 36: Inventories increased MoM

Japan laser diode inventory volume and months of inventory (mn units)



Source: METI (ministry of economy trade and industry)

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Exhibit 37: Stocks mentioned in this report

Current prices (8/19 close) and investment opinions of the stocks mentioned in this report

BofA ticker	TSE code	Company	Price	QRQ
NBCLF	3110 JP	Nittobo	¥ 2,996	C-1-7
AHKSF	3407 JP	Asahi Kasei	¥ 1,617	B-1-7
SUMCF	3436 JP	SUMCO	¥ 3,517	C-1-7
XZJCF	5706 JP	Mitsui Kinzoku	¥ 27,505	C-1-7
SHECF	4063 JP	Shin-Etsu Chem	¥ 6,090	B-1-7
FUWAF	5801 JP	Furukawa Electric	¥ 3,770	C-1-7
FKURF	5803 JP	Fujikura	¥ 5,328	C-1-7

Source: BofA Global Research

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Price objective basis & risk

Nittobo (3110 / NBCLF)

The price objective of ¥5,480 is derived by applying a 38x P/E multiple to FY3/28 earnings forecasts, using the upper bound of the historical P/E range of 4-38x. Downside risks to the share price are (1) a decline in sales of Nittobo's glass fiber products due to slowdowns in the electronics industry, such as for smartphones, base stations, or servers, (2) sharp yen appreciation (each \ 1 appreciation against the dollar erodes annual OP by an estimated \190mn), and (3) sharp rises in crude oil prices (each \$1/bbl increase in the crude price erodes annual OP by an estimated \ 80mn). Upside risks are (1) stronger-than-expected momentum in the electronics industry, (2) sharp yen depreciation, and (3) sharp declines in crude oil prices.

Asahi Kasei (3407 / AHKSF)

Our PO for Asahi Kasei is ¥1,960, based on sum-of-the-parts method with EV/EBITDA multiples on our FY3/28E implying EV/EBITDA multiple of 7.1x. Our EV/EBITDA multiples by segment are: 6.0x for materials, 7.0x for homes, and 8.0x for healthcare.

Upside risks are (1) brisk demand for petrochemicals, automobiles, electronics, and homes, (2) rapid growth in pharmaceuticals/medical device markets relevant to the company, (3) major restructuring in the petrochemicals business, and (4) substantial yen depreciation (each ¥1 depreciation boosts annual OP by ¥1.5bn against the dollar and by ¥0.3bn against the euro). Downside risks are (1) depressed petrochemicals prices, (2) deterioration of automotive/electronics markets, (3) declines in housing orders, (4) slower growth of pharmaceuticals/medical device markets relevant to the company, and (5) substantial yen appreciation.

SUMCO (3436 / SUMCF)

Our ¥6,500 price objective is based on a P/E of 22x our FY12/28 EPS estimate. This remains below the midpoint of the historical valuation range of 6-47x. Given our expectation for steady earnings growth through FY12/29, we believe a valuation in line with a normal-cycle multiple is justified.

Downside risks to our PO are (1) sharp yen appreciation (¥1/\$ appreciation reduces annual OP by ¥1.2bn), (2) an abrupt slowdown in the semiconductor market, and (3) substantial polysilicon inventory write-downs arising from losses at the gross profit level.

Resonac Holdings (4004 / SHWDF)

The price objective is ¥22,500. The FY12/27 P/E is 30x, a level higher than the midpoint of the 10-42x range seen after IFRS adoption in 2023. As the expansion of the artificial-intelligence industry becomes more pronounced, demand for AI server-related materials is expected to be stronger and more sustained, and therefore we apply a higher valuation level

Upside risks are (1) sale of the graphite electrodes business, along with partial spin-offs of low earners such as petrochemicals, and (2) high HBM growth at semiconductor majors among Resonac customers. Downside risks are (1) a sharp fall in AI-related materials sales on technological advances such as hybrid bonding, (2) large write-downs, and (3) equity financing. Also, while the dropout of goodwill amortization with IFRS (¥17.2bn annually) is a boost for nominal EPS we note the risk of large impairment losses on businesses with low profits

Shin-Etsu Chem (4063 / SHECF)

Our price objective is ¥7,900. This is derived by multiplying our FY3/28 EPS estimate by a P/E of 23x, near the upper limit of last 10 year range, 8-28x. We think this level is appropriate in light of stronger growth potential in earnings. Downside risks are a sudden decline in demand for semiconductors, a sudden decline in global demand for



PVC resin, and a rapid appreciation of the yen (each ¥1/USD appreciation depresses annual RP by ¥3.7bn). Upside risks are a sharp rise in semiconductor demand, sharp increase in global PVC resin demand and a weaker JPY.

Furukawa Electric (5801 / FUWAF)

Our price objective (PO) for Furukawa Electric is ¥7,300, based on a P/E of 40.0x our FY3/28 EPS estimate. We use a level over the midpoint of the historical 6-45x range over the past ten years, incorporating the structural earnings uplift driven by high growth in AI server-related materials.

Upside risks to the price objective are: (1) a sharp increase in data center investment, and (2) rapid yen depreciation (a ¥1/\$ depreciation would increase annual OP by ¥700mn).

Downside risks are: (1) intensifying competition and share loss in thermal components due to technological changes, (2) a sharp decline in data center investment, (3) rapid yen appreciation, and (4) the execution of equity financing.

Fujikura (5803 / FKURF)

Our PO of ¥8,000 is based on a P/E of 40.0x our FY3/28 EPS estimate. We use a multiple higher than the median of Fujikura's P/E range of 5-55x as it benefits significantly from structural growth for AI data centers.

Downside risks to our PO are (1) rapid yen appreciation (¥1 rise vs USD depresses annual OP by ¥2.0bn), (2) a sharp decline in optical communications demand, and (3) a swift slowdown in the smartphone and/or PC market.

Upside risks are (1) rapid yen depreciation, (2) sharp growth in optical communications demand, and (3) stronger-than-expected smartphone, PC market, and other IT-related demand.

Analyst Certification

I, Takashi Enomoto, hereby certify that the views expressed in this research report accurately reflect my personal views about the subject securities and issuers. I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

Japan - Cyclical Materials Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Adeka Corp	ADKCF	4401 JP	Niima Komura
	AGC Inc	ASGLF	5201 JP	Kyoji Chiba
	Asahi Kasei	AHKSF	3407 JP	Takashi Enomoto
	Denka	DENKF	4061 JP	Takashi Enomoto
	ENEOS Holdings	JXHGF	5020 JP	Takashi Enomoto
	Fujikura	FKURF	5803 JP	Takashi Enomoto
	Fujimi Incorporated	FUJXF	5384 JP	Niima Komura
	Furukawa Electric	FUWAF	5801 JP	Takashi Enomoto
	Fuso Chemical	XQRDF	4368 JP	Niima Komura
	Idemitsu Kosan	IDKOF	5019 JP	Takashi Enomoto
	JX Advanced Metals Corporation	XJTIF	5016 JP	Niima Komura
	Kansai Paint	KSANF	4613 JP	Takashi Enomoto
	Marubeni	MARUF	8002 JP	Takashi Enomoto
	Marubeni	MARUY	MARUY US	Takashi Enomoto
	MARUWA	MAWAF	5344 JP	Niima Komura
	Mitsubishi Corp.	MSBHF	8058 JP	Takashi Enomoto
	Mitsubishi Gas Chemical	MBGCF	4182 JP	Takashi Enomoto
	Mitsubishi Materials	MIMTF	5711 JP	Takashi Enomoto
	Mitsui & Co.	MITSF	8031 JP	Takashi Enomoto
	Mitsui & Co.	MITSY	MITSY US	Takashi Enomoto
	Mitsui Kinzoku	XZJCF	5706 JP	Takashi Enomoto
	MODEC	MDIKF	6269 JP	Takashi Enomoto
	Nichias	NICFF	5393 JP	Kyoji Chiba
	Nippon Electric Glass	NPEGF	5214 JP	Kyoji Chiba
	Nippon Sanso Holdings	XCIDF	4091 JP	Takashi Enomoto
	Nippon Steel	NISTF	5401 JP	Takashi Enomoto
	Nissan Chemical	NNCHF	4021 JP	Takashi Enomoto
	Nittobo	NBCLF	3110 JP	Takashi Enomoto
	NOF	NOFCF	4403 JP	Takashi Enomoto
	Osaka Soda	DACLF	4046 JP	Takashi Enomoto
	Rengo	RNGOF	3941 JP	Kyoji Chiba
	Resonac Holdings	SHWDF	4004 JP	Takashi Enomoto
	Sekisui Chemical	SKSUF	4204 JP	Takashi Enomoto
	Shin-Etsu Chem	SHECF	4063 JP	Takashi Enomoto
	SUMCO	SUMCF	3436 JP	Takashi Enomoto
	Sumitomo Bakelite	SBKLF	4203 JP	Niima Komura
	Sumitomo Chem.	SOMMF	4005 JP	Takashi Enomoto
	Sumitomo Corp.	SSUMF	8053 JP	Takashi Enomoto
	Sumitomo Corp.	SSUMY	SSUMY US	Takashi Enomoto
	Sumitomo Electric Industries	SMTOF	5802 JP	Takashi Enomoto
	Sumitomo Metal Mining	STMNF	5713 JP	Takashi Enomoto
	Tokai Carbon	TKCBF	5301 JP	Takashi Enomoto
	Tokyo Ohka Kogyo	TOKCF	4186 JP	Niima Komura
	Toyo Gosei	TYGIF	4970 JP	Niima Komura
	Tri Chemical Laboratories	XWURF	4369 JP	Niima Komura
NEUTRAL				
	Cosmo Energy Holdings	CEHCF	5021 JP	Takashi Enomoto
	INPEX	IPXHF	1605 JP	Takashi Enomoto
	Kaneka	KANKF	4118 JP	Kyoji Chiba
	Kumiai Chemical Industry	XHEXL	4996 JP	Kyoji Chiba
	NGK Corporation	NGKIF	5333 JP	Kyoji Chiba
	Nippon Paint Holdings	NPCPF	4612 JP	Takashi Enomoto
	Nippon Paper Industries	NJPF	3863 JP	Kyoji Chiba
	Nippon Soda	NSDFF	4041 JP	Kyoji Chiba
	Oji Holdings	OJIPF	3861 JP	Kyoji Chiba
	Osaka Organic Chemical	XAPKF	4187 JP	Niima Komura
	Sumitomo Osaka Cement	SUCEF	5232 JP	Kyoji Chiba
	Tekscend Photomask Corp.	XJXHP	429A JP	Niima Komura
	Toyota Tsusho	TYHOF	8015 JP	Takashi Enomoto
UNDERPERFORM				
	Air Water	AWTRF	4088 JP	Takashi Enomoto
	Daicel Corp	DACHF	4202 JP	Takashi Enomoto
	Iwatani Corporation	IWTNF	8088 JP	Takashi Enomoto



Japan - Cyclical Materials Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	JFE Holdings	JFEEF	5411 JP	Takashi Enomoto
	Kobe Steel	KBSTF	5406 JP	Takashi Enomoto
	Kuraray	KURRF	3405 JP	Takashi Enomoto
	Kureha	KURCF	4023 JP	Niima Komura
	Mitsubishi Chemical Group	MTLHF	4188 JP	Takashi Enomoto
	Mitsui Chemicals	MITUF	4183 JP	Takashi Enomoto
	Nippon Shokubai	NSHKF	4114 JP	Takashi Enomoto
	Taiheiyō Cement	THYCF	5233 JP	Kyoji Chiba
	Toray	TRYIF	3402 JP	Takashi Enomoto
	Tosoh Corporation	TOSCF	4042 JP	Takashi Enomoto
	Zeon	ZEOOF	4205 JP	Niima Komura

Disclosures

Important Disclosures

Equity Investment Rating Distribution: Chemicals Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	67	48.55%	Buy	29	43.28%
Hold	29	21.01%	Hold	19	65.52%
Sell	42	30.43%	Sell	22	52.38%

Equity Investment Rating Distribution: Electronics Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	34	55.74%	Buy	16	47.06%
Hold	10	16.39%	Hold	3	30.00%
Sell	17	27.87%	Sell	6	35.29%

Equity Investment Rating Distribution: Non-Ferrous Metals/Mining & Minerals Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	65	63.73%	Buy	36	55.38%
Hold	21	20.59%	Hold	10	47.62%
Sell	16	15.69%	Sell	7	43.75%

Equity Investment Rating Distribution: Global Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	1987	56.11%	Buy	1190	59.89%
Hold	797	22.51%	Hold	496	62.23%
Sell	757	21.38%	Sell	391	51.65%

^{R1} Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

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Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster ^{R2}
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

^{R2} Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

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